CS 4530: Fundamentals of Software Engineering

Module 08: Concurrency Patterns in Typescript

Adeel Bhutta

Khoury College of Computer Sciences

Learning Goals for this Lesson

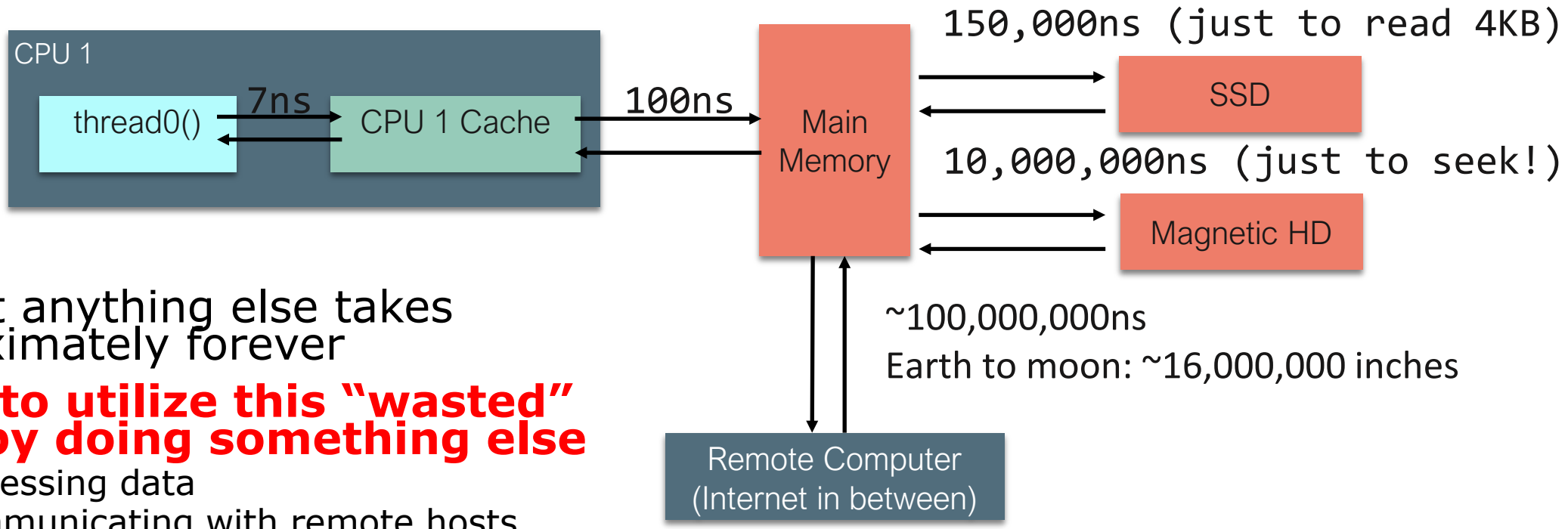
- At the end of this lesson, you should be prepared to:
  - Explain the difference between JS run-to-completion semantics and interrupt-based semantics.
  - Given a simple program using async/await, work out the possible orders in which the statements in the program will run.
  - Write simple programs that create and manage promises using async/await
  - Write simple programs to mask latency with concurrency by using non-blocking IO and Promise.all in TypeScript.

Why async matters

- Your browser probably spends 99% of its time waiting
  - for http requests to return
  - for database calls to return
- Wouldn't it be nice if we could use that waiting time for something else?
  - maybe we have some other http request we could launch
  - maybe other things (throbbers??)
- **async** is Typescript's way of handling this

Your app probably spends most of its time waiting

- Consider: a 1Ghz CPU executes an instruction every 1 ns



- Almost anything else takes approximately forever
- **Want to utilize this "wasted" time by doing something else**
  - Processing data
  - Communicating with remote hosts
  - Timers that countdown while our app is running
  - Echoing user input

Typescript/Javascript achieves this goal using two techniques:

1. cooperative multiprocessing

2. non-blocking IO

Most OS's use **pre-emptive multiprocessing**

- OS manages multiprocessing with multiple threads of execution
- Processes may be ***interrupted*** at unpredictable times
- Inter-process communication by shared memory
- Data races abound
- Really, really hard to get right: need critical sections, semaphores, monitors (all that stuff you learned about in op. sys.)

Javascript/Typescript uses **cooperative multiprocessing**

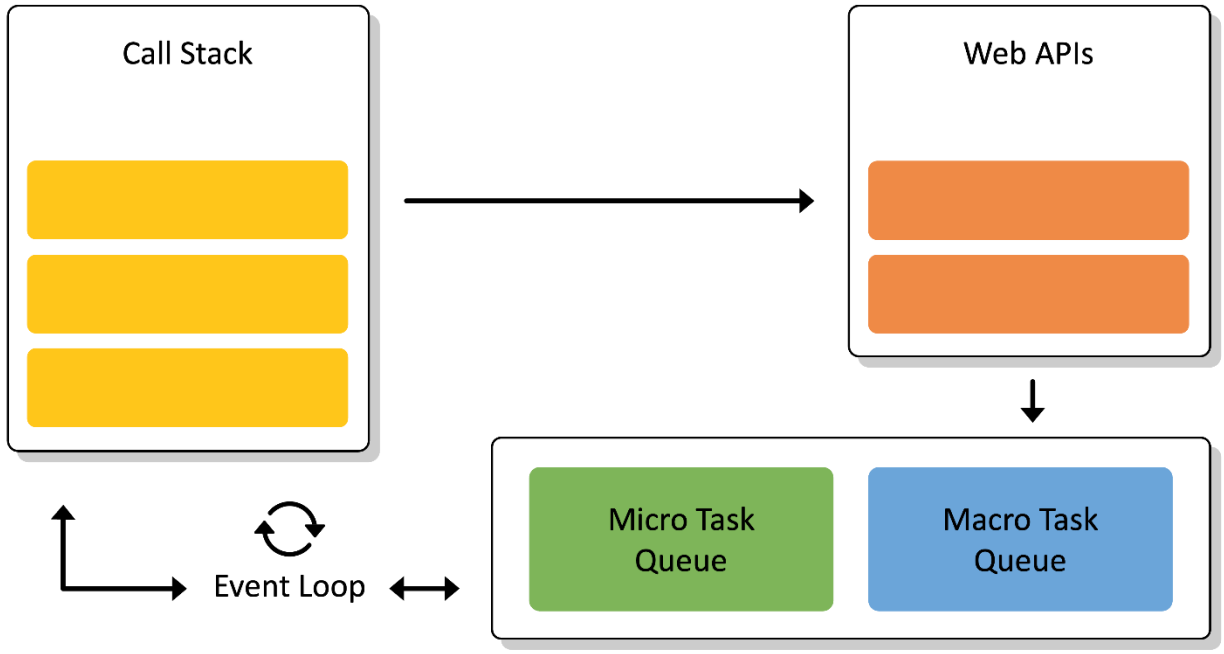
- In cooperative multiprocessing, only one process is executed at a time.
- Each process pauses when it is convenient to allow other processes to make progress.
- To make this practical (and avoid cheating!), we need a programming model that encourages this behavior

async/await is a programming model for cooperative multiprocessing

- In async/await, the program is organized into a set of "async functions"
- An async function is like an ordinary function, except that it may be executed by webAPI or the browser
- If you **await** on async function, it can pause the execution until async function is completed. This is also known as a 'promise'. We will show you those with an example.
- When one program pauses, the runtime can choose to resume executing any process that is ready to run.

How does JS Event Loop make this happen?

- One Event Loop means that we have single thread of execution
- WebAPI are often used for asynchronous tasks
- Queues are used for “await”-ing tasks
- When call stack gets empty, event loop picks up tasks from Callback Queues
- Microtask queue has priority over macrotask queue



Run to Completion is a central concept

- In JS, you work on one and one task only (i.e., event loop has a single call stack)
- Also, if you start a task, you must finish it (i.e., call stack must be emptied) before you start another task. This pattern is called "run-to-completion" semantics
- A pause point will correspond to each *await* keyword which allows you to work on a different unit of work.
- You can do lots of different things with promises.
- Let's look some typical patterns.

A typical async function

```
async function someFunction(i: number) {
    const j = i + 1;
    // ...
    const k = await someOtherAsyncFunction(j);
    // ...
    const m = k + 100
    return m;
}
```

Execution of an async function may be paused

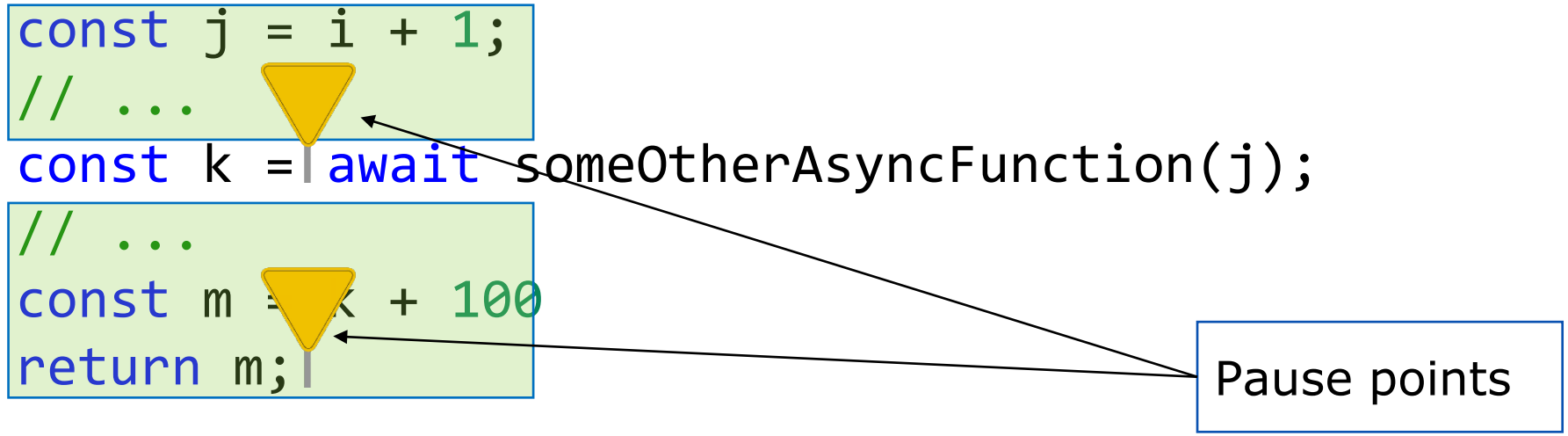
```
async function someFunction(i: number) {
    const j = i + 1;
    // ...
    const k = await someOtherAsyncFunction(j);
    // ...
    const m = k + 100;
    return m;
}
```

It will never pause in here. This is called a Critical Section

Or in here (this is another Critical Section)

What happens at those pause points?

```
async function someFunction(i: number) {
    const j = i + 1;
    // ...
    const k = await someOtherAsyncFunction(j);
    // ...
    const m = k + 100
    return m;
}
```



What happens when you await?

- When you ‘await’ on an asynchronous task, you are cooperating with the event loop by allowing it to work on something else.
- In technical terms, you are returning with a ***pending*** promise which means that some work will be completed in future.
- While you are waiting for that future, event loop will continue finishing its current task (whatever is in call stack) and then other tasks (whatever is in callback queue).
- Eventually you will return to complete the ***awaited*** work
- Let’s understand those with some examples

src/Slides/dataRace2.ts

Java vs. JS/TS

```
async function asyncPrintA() {
    await waitRandom(1);
    console.log('A');
}

async function asyncPrintBC() {
    await waitRandom(2);
    console.log('B');
    console.log('C');
}

async function run() {
    await Promise.all([asyncPrintA(), asyncPrintBC()])
}

run()
```

- In Java, you could get an interrupt between the print B and the print C.
- So the output could be ABC, BAC, BCA
- In TS/JS, the print B and print C are in the same critical section, so BAC is impossible!

A promise can be in one of exactly 3 states

- A JavaScript promise can be in one of three states: pending, fulfilled, or rejected.
- Pending is the initial state where the promise is ***waiting*** for an operation to complete; Then it becomes Ready for Execution; and eventually it is Executing before being resolved
- Resolved: either fulfilled or rejected.
  - fulfilled means the operation was successful,
  - rejected indicates that the operation failed.

src/Slides/newRequest.ts

Simple sequential execution:

```
// simple synchronous execution
async function main() {
    console.log('main started');
    console.log(`sync task`);
    console.log('main done');
}

main()
```

- Everything runs sequentially

```
$ node newRequest.ts
main started
Sync task
main done
```

src/Slides/newRequest.ts

Async task can finish later

```
// simple synchronous execution
async function main() {
    console.log('main started');
    console.log(`sync task`);
    setTimeout(() => {
      console.log('timeout');
    }, 0);
    console.log('main done');
}

main()
```

- Run to completion requires main to finish first
- setTimeout was given to WebAPI

```
$ node newRequest.ts
main started
sync task
main done
timeout
```

src/Slides/newRequest.ts

Promise can also be resolved in future

```
// simple synchronous execution
async function main() {
    console.log('main started');
    console.log(`sync task`);
    Promise.resolve().then( () => {
      console.log('promise');
    } );
    console.log('main done');
}

main()
```

```
$ node newRequest.ts
main started
sync task
main done
promise
```

src/Slides/newRequest.ts

Promises have priority (Micro over Macro)

```
// simple synchronous execution
async function main() {
    console.log('main started');
    console.log(`sync task`);
    Promise.resolve().then( () => {
      console.log('promise');
    } );
    setTimeout(() => {
      console.log('timeout');
    }, 0);
   console.log('main done');
}

main()
```

```
$ node newRequest.ts
main started
sync task
main done
promise
timeout
```

src/Slides/oneRequest.ts

You can await on async tasks (1)

```
// fakeRequest(n) is an async that waits for 1 second and then
// resolves with the number n+10
import { fakeRequest } from "./fakeRequest";
import { timeIt } from "./timeIt";

async function main() {
    console.log('main started');
    const res = await fakeRequest(32);
    console.log(`fakeRequest(${request}) returned: ${res}`);
    console.log('main done');
}

timeIt(main)
```

```
export function fakeRequest(n: number): Promise<number> {
  return new Promise((resolve, reject) => {
    console.log('fakeRequest received request:', n);
    setTimeout(() => {
      console.log('time passes....');
      resolve(n + 10);
    }, 1000);
  });
}
```

```
$ node oneRequest.ts
main started
fakeRequest received request: 32
time passes....
fakeRequest(32) returned: 42
main done
1015.98 msec
```

src/Slides/oneRequest.ts

You can await on async tasks (2)

```
// makeRequest is a utility function to simulate API requests and
// then response
import { makeRequest } from "./fakeRequest";
import { timeIt } from "./timeIt";

async function main() {
    console.log('main started');
    await makeRequest(12);
    console.log('main done');
}

timeIt(main)
```

```
export async function makeRequest(requestNumber:number) {
    console.log(`starting makeRequest(${requestNumber})`);
    const response = await fetch('https://rest-example.covey.town');
    console.log('request:', requestNumber, '; response:', response.data);
}
```

```
$ node oneRequest.ts
main started
starting makeRequest(12)
request: 12
response: This is GET number 190 on
the current server
main done
264.65 msec
```

src/Slides/makeOneRequests.ts

Pattern for starting a concurrent computation using non-blocking I/O

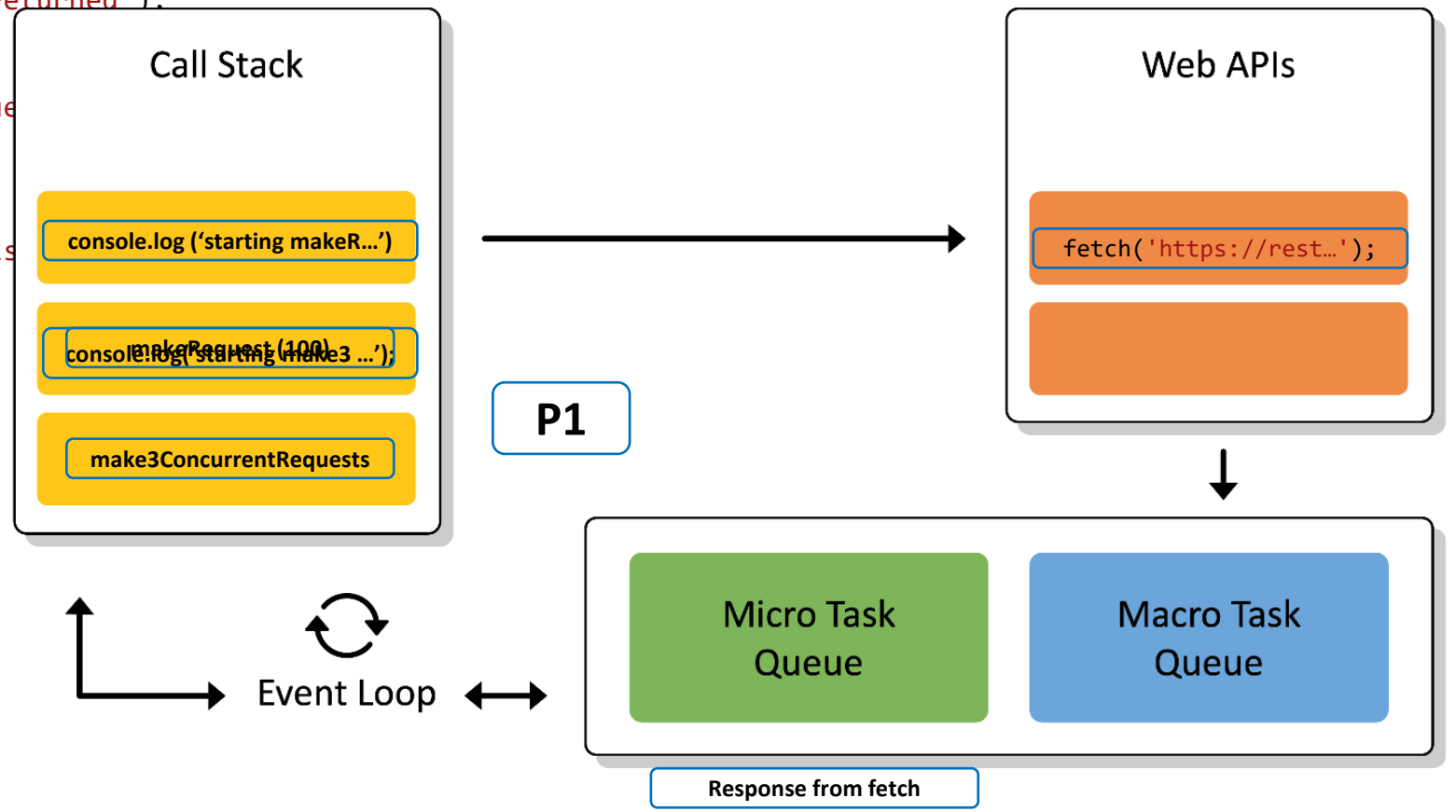
```
export async function makeRequest(requestNumber:number) {
    console.log(`starting makeRequest(${requestNumber})`);
    const response = await fetch('https://rest-example.covey.town');
    console.log('request:', requestNumber, '; response:', response.data);
}
```

1. The first console.log is printed
2. The http request is *immediately* sent, using non-blocking i/o. The browser goes about its business
3. A pending promise is returned to the caller. Caller can continue its other business.
4. Eventually, the fetch returns.
5. Some time after that, the console.log is printed and the makeRequest concludes.
6. Any promises that are waiting for the result of this makeRequest become eligible for execution.

src/Slides/makeThreeConcurrentRequests.ts

Running 3 concurrent fetch requests (start)

```
import axios from 'axios';
export async function makeRequest(requestNumber:number) {
    console.log(`starting makeRequest(${requestNumber})`);
    const response = await fetch('https://rest-example.covey.town');
    console.log(`request:${requestNumber} returned`);
}
function make3ConcurrentRequests() {
  console.log('starting make3ConcurrentReque
  makeRequest(100);
  makeRequest(200);
  makeRequest(300);
  console.log('make3ConcurrentRequests finis
}
make3ConcurrentRequests()
```

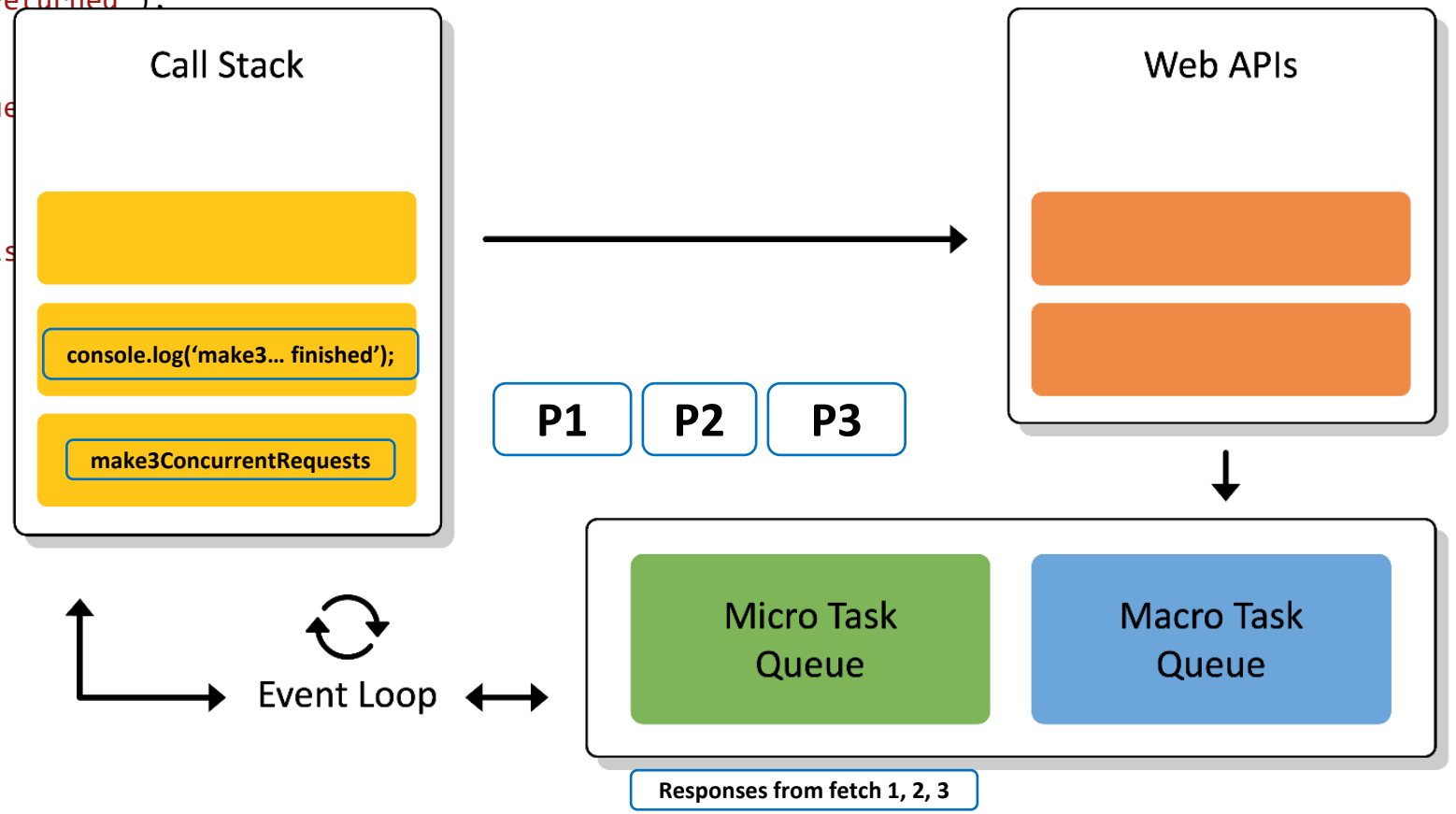


- Now, makeRequest(200) can proceed …

src/Slides/makeThreeConcurrentRequests.ts

Running 3 concurrent fetch requests (end)

```
import axios from 'axios';
export async function makeRequest(requestNumber:number) {
    console.log(`starting makeRequest(${requestNumber})`);
    const response = await fetch('https://rest-example.covey.town');
    console.log(`request:${requestNumber} returned`);
}
function make3ConcurrentRequests() {
  console.log('starting make3ConcurrentReque
  makeRequest(100);
  makeRequest(200);
  makeRequest(300);
  console.log('make3ConcurrentRequests finis
}
make3ConcurrentRequests()
```



src/Slides/makeThreeConcurrentRequests.ts

Running 3 concurrent fetch requests

```
import axios from 'axios';

export async function makeReq
    console.log(`starting mak
    const response = await fe
    console.log(`request:${re
}


  function make3ConcurrentReq
    console.log('starting mak
    makeRequest(100);
    makeRequest(200);
    makeRequest(300);
    console.log('make3ConcurrentRequests finished');
  }

make3ConcurrentRequests()
```

```
$ node makeThreeConcurrentRequests.ts
starting make3ConcurrentRequests
starting makeRequest(100)
starting makeRequest(200)
starting makeRequest(300)
make3ConcurrentRequests finished
request 300 returned
request 100 returned
request 200 returned
```

- 3 requests may finish in any order

If you add awaits, the requests will be processed sequentially

```
async function main() {
    console.log('starting main');
    const res1 = await fakeRequest(1);
    console.log(`fakeRequest(1) returned: ${res1}`);
    const res2 = await fakeRequest(2);
    console.log(`fakeRequest(2) returned: ${res2}`);
    const res3 = await fakeRequest(2);
    console.log(`fakeRequest(2) returned: ${res3}`);
    console.log('main done');


}

timeIt(main)
```

```
$ node threeRequestsSequentially.ts
starting main
fakeRequest received request: 1
time passes....
fakeRequest(1) returned: 11
fakeRequest received request: 2
time passes....
fakeRequest(2) returned: 12
fakeRequest received request: 3
time passes....
fakeRequest(3) returned: 13
main done
3024.03 msec
```

src/Slides/threeRequestsSequentially.ts

Use Promise.all to execute several requests concurrently

src/Slides/threeRequestsConcurrently.ts

```
async function main() {
    console.log('starting main');
    const promises = [fakeRequest(1),
                      fakeRequest(2),
                      fakeRequest(3)];
    const results = await Promise.all(promises);
    console.log('results:', results);
    console.log('main done');
}

timeIt(main)
```

```
$ node threeRequestsConcurrently.ts
starting main
fakeRequest received request: 1
fakeRequest received request: 2
fakeRequest received request: 3
time passes....
time passes....
time passes....
results: [ 11, 12, 13 ]
main done
1018.81 msecno
```

- 3 requests may finish in any order

Why is that?
Visualizing Promise.all

Sequential (await)

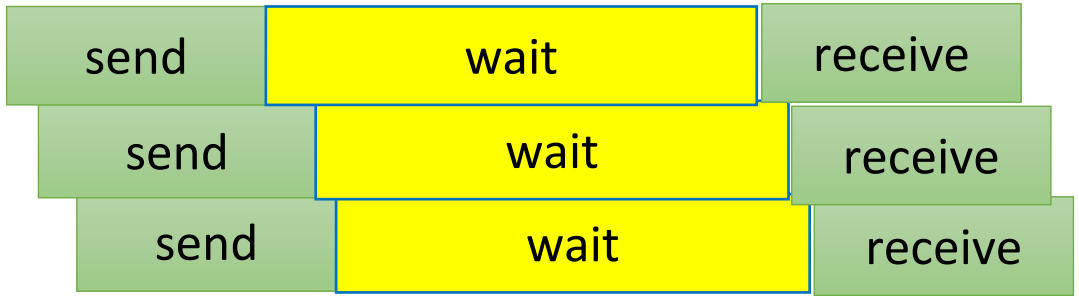
"Don't make another request until you got the last response back"

237 msec

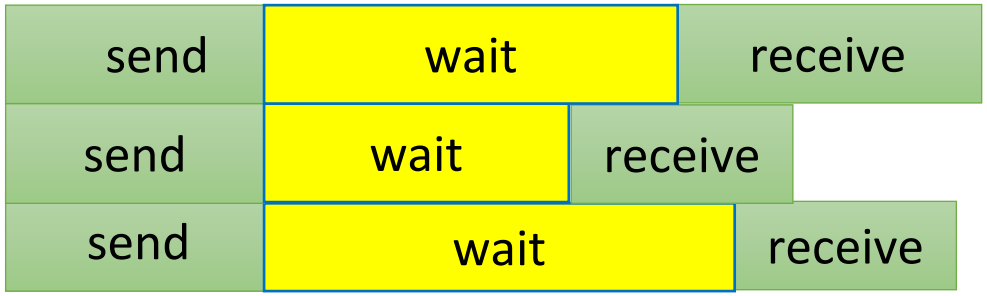
Concurrent (Promise.all)

"Make all of the requests now, then wait for all of the responses"

34 msec



or



Requests can also be chained (if they are serial)

```
async function main() {
    console.log('main started');
    const request1 = 32
    const res1 = await fakeRequest(request1);
    console.log(`fakeRequest(${request1}) returned: ${res
    const res2 = await fakeRequest(res1);
    console.log(`fakeRequest(${res1})returned: ${res2}`);
    const res3 = await fakeRequest(res2);
    console.log(`fakeRequest(${res2})returned: ${res3}`);
    console.log([request1, res1, res2, res3]);
    console.log('main done');
}
```

src/Slides/threeRequestsChained.ts

```
$ node threeRequestsChained.ts
main started
fakeRequest received request: 32
time passes....
fakeRequest(32) returned: 42
fakeRequest received request: 42
time passes....
fakeRequest(42)returned: 52
fakeRequest received request: 52
time passes....
fakeRequest(52)returned: 62
[ 32, 42, 52, 62 ]
main done
3080.99 msec
```

Recover from errors with try/catch

```
// a request that may fail
async function maybeFailingRequest(req: number): Promise<number> {
    const res = await fakeRequest(req);
    if (res < 0) {
        throw new Error(`Request ${req} failed because response ${res} < 0`);
    } else {
        return res;
    }
}
```

src/Slides/tryCatchExample.ts

try/catch, continued

```
async function main() {
    console.log('main started');
    const req1 = -32
    let res: number;
    try {
        res = await maybeFailingRequest(req1);
        console.log(`fakeRequest(${req1}) returned: ${res}`);
    } catch (err) {
        console.error(`Error occurred for request ${req1}`);
        res = 0
    }
    console.log('main done with res =',
}

timeIt(main);
```

```
$ node tryCatchExample.ts
main started
fakeRequest received request: -32
time passes....
Error occurred for request -32
main done with res = 0
1007.52 msec
```

Pattern for testing an async function

src/Slides/example.test.ts

```
import { fakeRequest } from './fakeRequest.ts'
import { describe, expect, test} from 'vitest'

// fakeRequest should make a network request using myFetch
// and then returns n + 1

describe('fakeRequest', () => {

    test('fakeRequest should return its argument + 1 (await/expect)', async () => {
        expect.assertions(1)
        await expect(fakeRequest(100)).resolves.toEqual(101)
    })

    test('fakeRequest should return its argument + 1 (expect/await)', async () => {
        expect.assertions(1)
        expect(await fakeRequest(7)).toEqual(8)
    })
```

src/Slides/oneRequestNoAwait.ts

AntiPattern 1: un**await**ed promise

```
// fakeRequest(n) is an async that waits for 1 second and then
// resolves with the number n+10
import { fakeRequest } from "./fakeRequest";
import { timeIt } from "./timeIt";

async function main() {
    console.log('main started');
    const request = 32
    const res = fakeRequest(request);
    console.log(`fakeRequest(${reques
    console.log('main done');
}

timeIt(main)
```

```
$ node oneRequestNoAwait.ts
main started
fakeRequest(32) returned: [object Promise]
main done
2.64 msec
fakeRequest received request: 32
time passes....
```

AntiPattern 2: Side-effect before **await** can be misleading

src/Slides/antiPattern2.ts

```
async function f() {
    console.log('f started');
    await g();
    console.log('f done');
}

async function g() {
    console.log('g started');
    const res = await fakeRequest(32);
    console.log(`fakeRequest(32) returned:
${res}`);
}
```

Critical Section #1: no interruption between 'f started' and 'g started'

Critical Section #2

Critical Section #3

But where does the non-blocking IO come from?

We achieve this goal using two techniques:

1. cooperative multiprocessing

2. non-blocking IO 

Answer: JS/TS has some primitives for starting a non-blocking computation

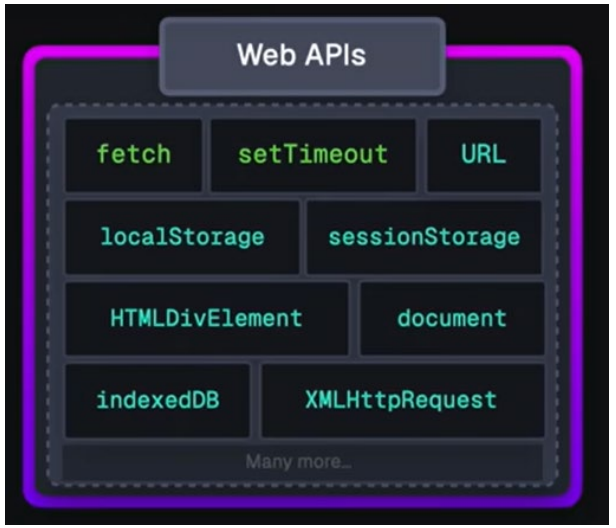
- These are things like http requests, I/O operations, or timers.
- Each of these returns a promise that you can **await**. The promise runs while it is pending, and produces the response from the http request, or the contents of the file, etc.
- You may use some of these directly, like **fetch** or **setTimeout**.
- More frequently, you will invoke these using a convenient library, like

```
axios.get('https://rest-example.covey.town’)
```

to make an http request, or

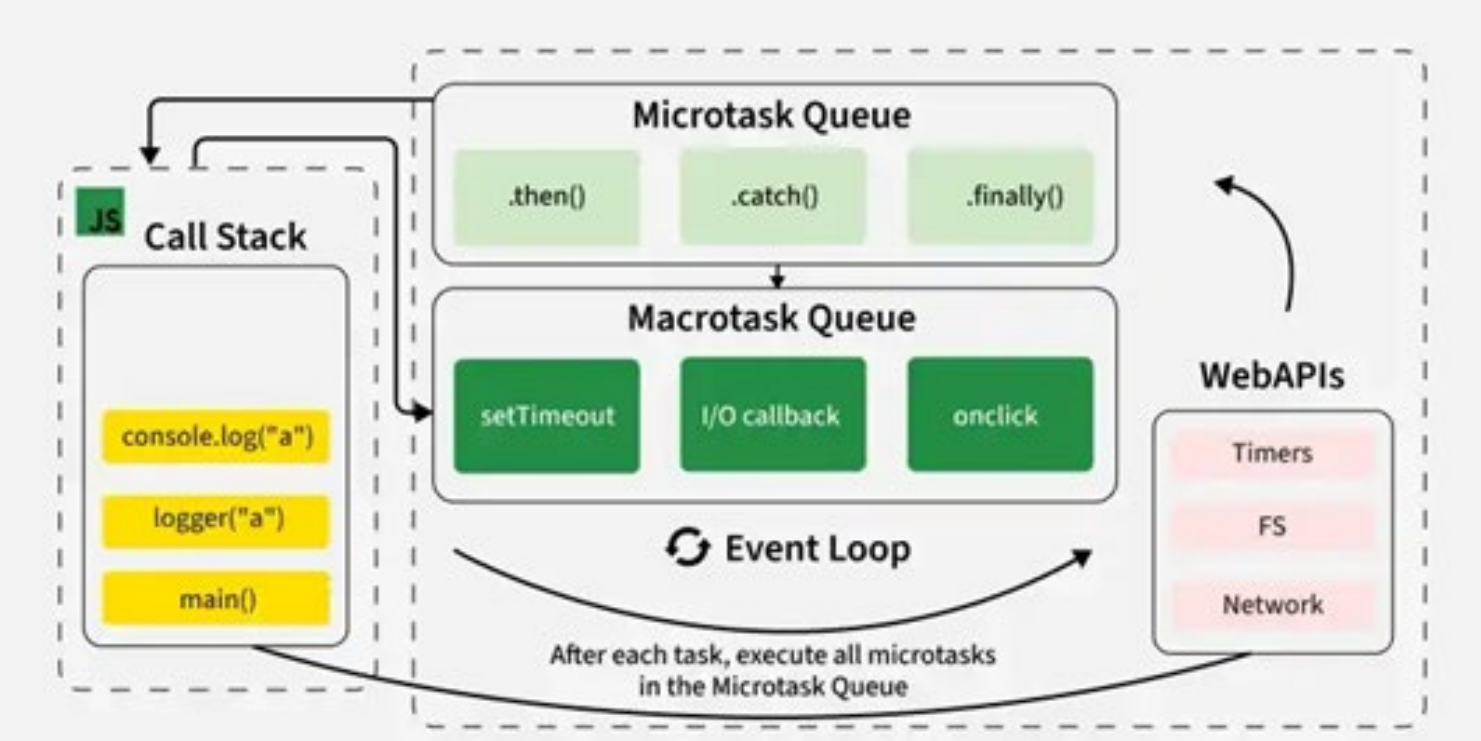
```
fs.readFile(filename)
```

to read the contents of a file.



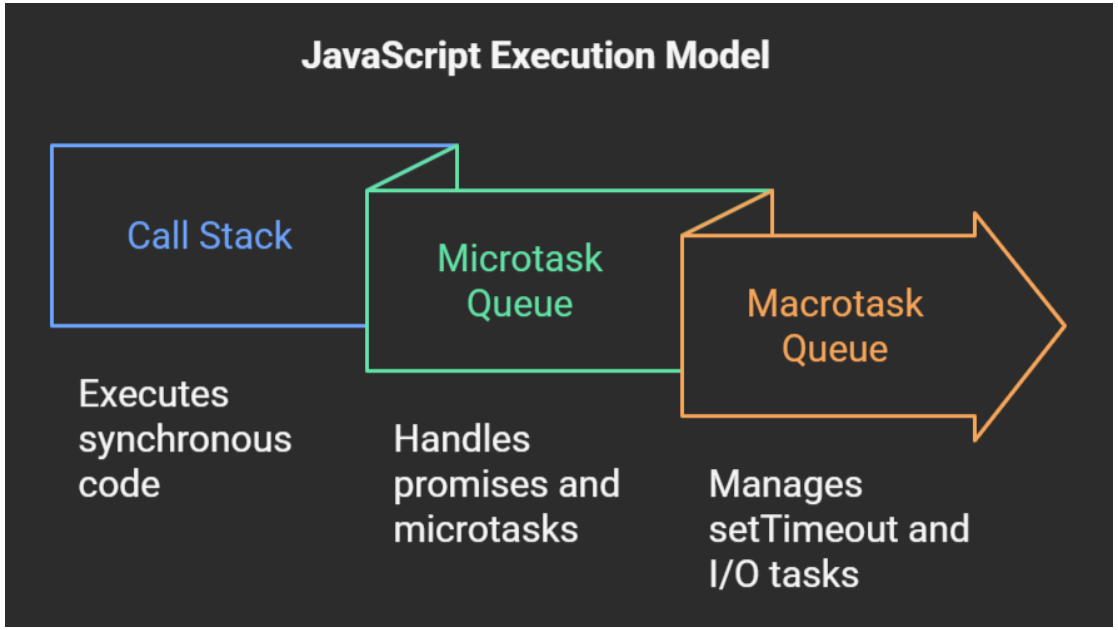
Let’s put it all together

- JS/TS has single event loop
- We outsource most of the work to WebAPIs for asynchronous work
- Upon completion, they are placed in queues.
- Event loop picks them up from queue when call stack is empty!



Microtasks are urgent and Macrotasks are slower/asynchronous operations

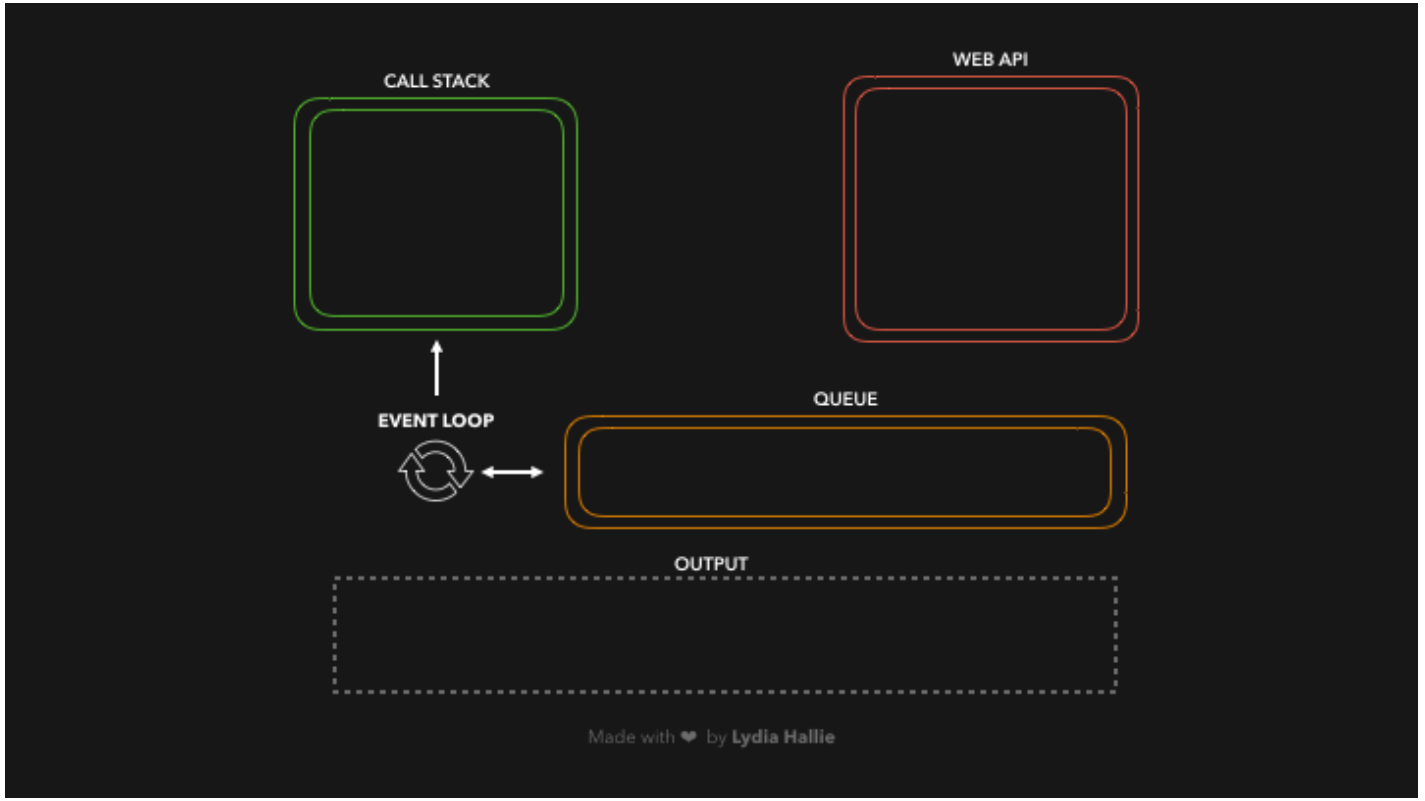
- Microtask queue is used for immediate/urgent tasks while Macrotask queue is used for slower asynchronous tasks.
- When call stack is empty, microtask queue is processed first (before browser renders). When microtask queue is empty then macrotask queue is processed.



Here is a quick demo for you

```
const foo = () => console.log("First");
const bar = () => setTimeout(() => console.log("Second"), 500);
const baz = () => console.log("Third");

bar();
foo();
baz();
```



Courtesy of https://dev.to/lydiahallie/javascript-visualized-event-loop-3dif

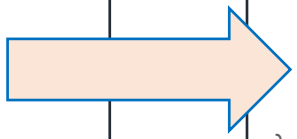
General Rules for Writing Asynchronous Code

- You can’t return a value from a promise to an ordinary procedure.
  - You can only send the value to another promise that is awaiting it.
- Call async procedures only from other async functions or from the top level.
- Break up any long-running computation into **async/await** segments so other processes will have a chance to run.
- Leverage concurrency when possible
  - Use **promise.all** if you need to wait for multiple promises to return (or **Promise.allSettled**, if needed).
- Check for errors with **try/catch**

Odds and Ends You Should Know

Async/await code is compiled into promise/then code. Using await is preferred

```
async function
makeThreeSerialRequests(){
1.   console.log('Making first
request');
2.   await makeOneGetRequest();
3.   console.log('Making second
request');
4.   await makeOneGetRequest();
5.   console.log('Making third
request');
6.   await makeOneGetRequest();
7.   console.log('All done!');
}
makeThreeSerialRequests();
```



```
console.log('Making first request');
makeOneGetRequest().then( () =>{
  console.log('Making second request');
  return makeOneGetRequest();
}).then(() => {
  console.log('Making third request');
  return makeOneGetRequest();
}).then(()=>{
  console.log('All done!');
});
```

src/Slides/dataRace.ts

async doesn't eliminate data races

```
import { myFetch } from './fakeRequest.ts';

const log: string[] = [];

async function X(n: number) {
    await myFetch(1)
    log.push('X');
}

async function Y(n: number) {
    await myFetch(2)
    log.push('Y');
}
for (let i = 0; i < 4; i += 1) {
    log.push('start');
    await Promise.all([X(10), Y(20)]);
}
console.log(log);
```

```
$ node dataRace.ts
Promise created for 1
Promise created for 2
Promise created for 1
Promise created for 2
Promise created for 1
Promise created for 2
Promise created for 1
Promise created for 2
[
  'start', 'X', 'Y',
  'start', 'Y', 'X',
  'start', 'Y', 'X',
  'start', 'Y', 'X'
]
```

Review

- You should now be prepared to:
  - Explain the difference between JS run-to-completion semantics and interrupt-based semantics.
  - Given a simple program using async/await, work out the possible orders in which the statements in the program will run.
  - Write simple programs that create and manage promises using async/await
  - Write simple programs to mask latency with concurrency by using non-blocking IO and Promise.all in TypeScript.